14. Common or Cotton cutworm (*Spodoptera litura*)

Last modified: October 16, 2026

14A. Introduction

The common or cotton cutworm (SLI), *Spodoptera litura* (Fabricius, 1775) (Lepidoptera: Noctuidae), is a highly polyphagous pest that feeds on at least 120 plant species, including economically important crops such as corn, cotton, groundnut, potato, soybean, sweet potato, tea, tobacco, and a wide range of vegetables (Venette et al. 2003, Sullivan 2007, Bragard et al. 2019, EPPO 2023). Females lay eggs on host plants, and within a few days, emerging larvae begin feeding on plant tissue, causing extensive damage that can lead to stunted growth and reduced marketability (Sullivan 2007). Native to Southeast Asia, SLI is now distributed across Australia, Oceania, several African islands, Hawaii, and other regions of Asia (Bragard et al. 2019, EPPO 2023). The pest has been present in Hawaii since 1964 and has been intercepted hundreds of times at United States ports of entry, primarily through cargo and passenger baggage (Venette et al. 2003, Sullivan 2007). The species has developed resistance to numerous insecticides and transgenic Bt cotton and is capable of long-distance migration, potentially aided by typhoons (Tu et al. 2010, Fu et al. 2015, Wan et al. 2008, Bragard et al. 2019, Murata et al. 1998). Severe defoliation caused by larval feeding has led to significant crop damage and economic losses in affected regions (Murata et al. 1998, Prasad et al. 2013, Bragard et al. 2019, Maharjan et al. 2023).

Potentially environmentally suitable areas for establishment for SLI in the contiguous United States (CONUS) include much of the southeastern United States and along the Pacific Coast (Coop 2018). If introduced to the United States, SLI could potentially overwinter in protected environments such as greenhouses or establish northern migratory populations (Bragard et al. 2019). Development and fecundity are strongly influenced by temperature, humidity, and host plant quality (Garad et al. 1985, EPPO 2023, Maharjan et al. 2023, Fu et al. 2025). SLI does not undergo diapause and therefore cannot survive winter conditions in areas north of approximately 30°N in China, where it instead establishes seasonal migratory populations (Fand et al. 2015, Fu et al. 2015, Bragard et al. 2019, EPPO 2023). SLI completes 3-12 generations in tropical regions versus three generations in colder parts of its distribution, including northern China (EPPO 2023). Consistent with its high reproductive potential under warm conditions, a DDRP model predicted one to ten possible generations annually in the conterminous United States, with four to six generations predicted across much of the Southeast and California (Coop 2018). SLI may also respond to changing temperatures through shifts in its geographic distribution and population growth, although warming could reduce suitability in some areas if temperatures exceed optimal conditions for development (Fand et al. 2015, Fu et al. 2025).

A model for SLI was developed for the Degree-Day, establishment Risk, and Phenological event mapping system (DDRP) is operationalized to produce near real-time forecasts for the CONUS at https://uspest.org/DDRP. The phenology model applies a lower developmental threshold of 10.6 ºC and upper threshold of 40 ºC for all life stages (egg, larva, pupa, and adult) and assumes the pupa is the overwintering stage. The climatic suitability model applies a lower cold stress threshold and upper heat stress threshold of −3 °C and 44 °C, respectively. The full set of model parameters for SLI, including degree-day requirements for individual life stages and climate stress limits, are presented in Table 1. Details on model parameterization are available at USPest.org (Coop 2018).

Table 1. SLI model parameters for DDRP.

Parameter	Code	Value
Lower developmental thresholds (°C)		

Parameter	Code	Value
Egg	eggLDT	10.56
Larvae	larvaeLDT	10.56
Pupae	pupaeLDT	10.56
Adult	adultLDT	10.56
Upper developmental thresholds (°C)		
Egg	eggUDT	40
Larvae	larvaeUDT	40
Pupae	pupaeUDT	40
Adult	adultUDT	40
Stage durations (°C degree-days)		
Egg	eggDD	56
Larvae	larvaeDD	288
Pupae	pupDD	165
Adult	adultDD	60
Pest events (°C degree-days)		
J-larval development complete	OWEventDD	120
Egg hatch	eggEventDD	5
Developing J-larvae	larvaeEventDD	144
Adult emergence (pupal development complete)	pupaeEventDD	165
Egg laying	adultEventDD	54
Cold stress		
Cold stress temperature threshold (°C)	coldstress_threshold	-3
Cold degree-day (°C) limit when most individuals die	coldstress_units_max1	25
Cold degree-day (°C) limit when all individuals die	coldstress_units_max2	425
Heat stress		
Heat stress temperature threshold (°C)	heatstress_threshold	44
Heat stress degree-day (°C) limit when most individuals die	heatstress_units_max1	75
Heat stress degree-day (°C) limit when all individuals die	heatstress_units_max2	150
Cohorts		
Degree-days (°C) to emergence (average)	distro_mean	110
Degree-days (°C) to emergence (variation)	distro_var	2000
Minimum degree-days (°C) to emergence	xdist1	80
Maximum degree-days (°C) to emergence	xdist2	210
Shape of the distribution	distro_shape	normal
Other		
Order of stages	stgorder	OP, A, E, L, P
Degree-day calculation method	calctype	triangle

14B. Results

i. Phenology

Entire period: 1981–2025

Over 1981–2025, predicted egg hatch generally shifted later across much of the northern and eastern United States, while adult emergence showed a broader mixture of earlier and later changes across the CONUS (Figure 1, Figure 2). Significant changes were concentrated in scattered portions of the eastern and central United States.

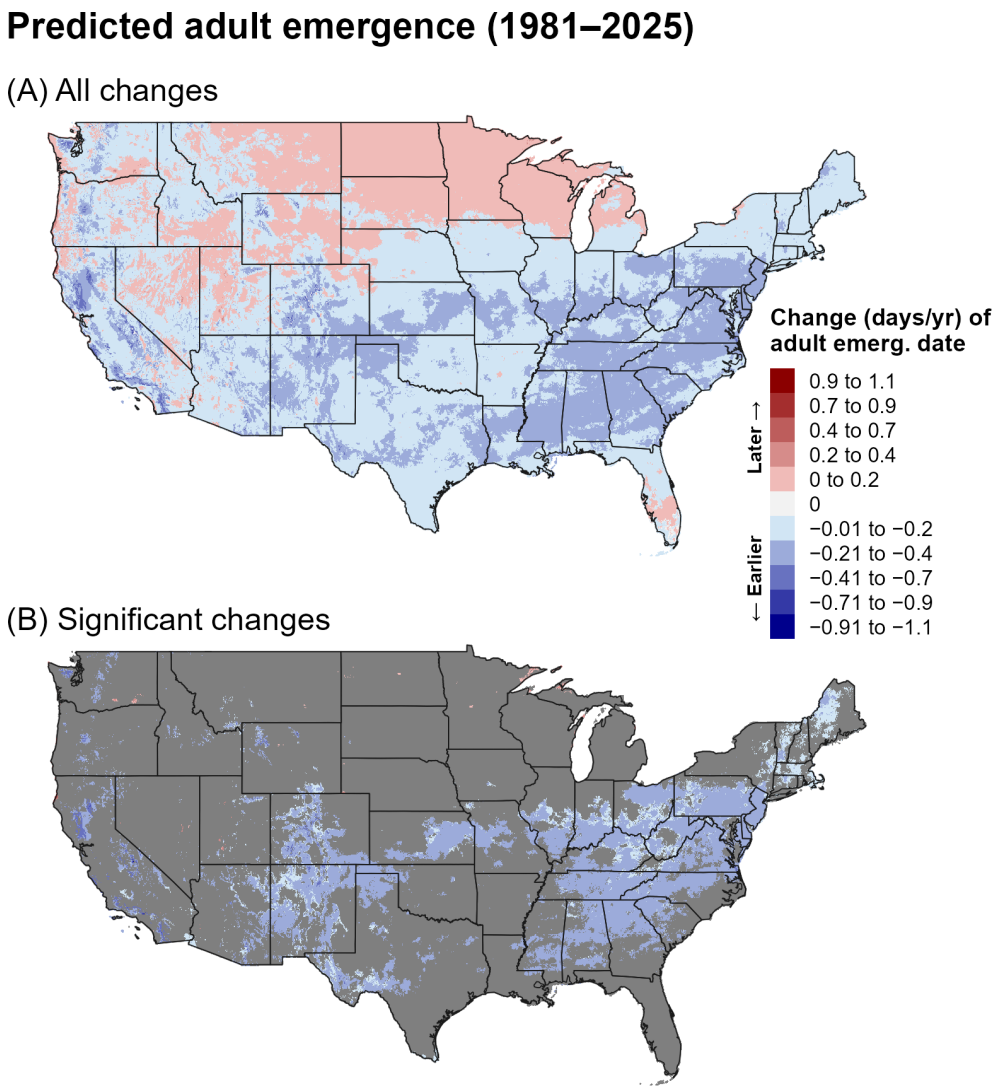


Figure 1. Changes (days/year) in predicted dates of first adult emergence for SLI across the CONUS over the entire 44-year period (1981–2025). Maps show (A) all changes and (B) statistically significant changes ($P < 0.1$); areas shown in gray are not statistically significant.

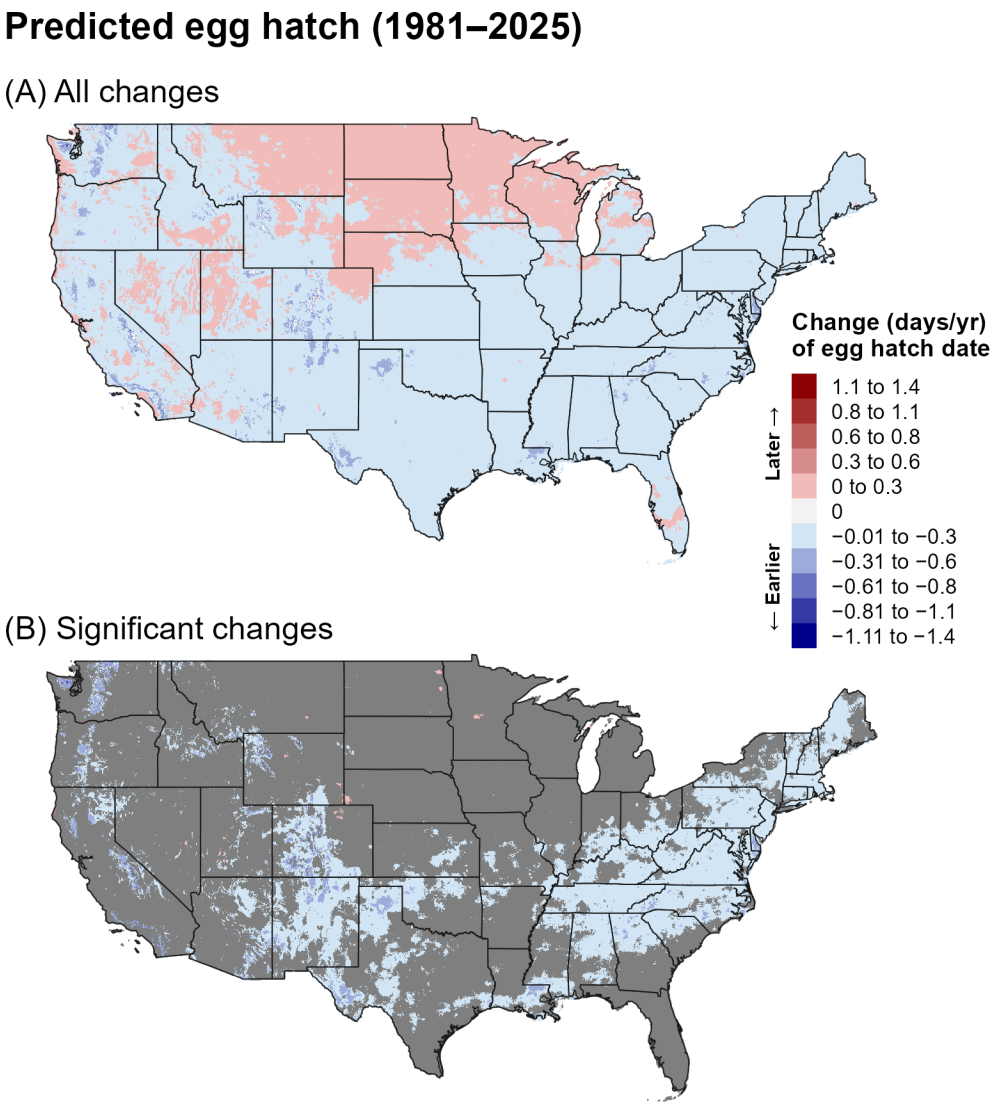


Figure 2. Changes (days/year) in predicted dates of first egg hatch for SLI across the CONUS over the entire 44-year period (1981–2025). Maps show (A) all changes and (B) statistically significant changes ($P < 0.1$); areas shown in gray are not statistically significant.

Recent period: 2001–2025

During 2001–2025, egg hatch showed earlier trends across much of the western and central United States, while adult emergence was later across portions of the northern interior and earlier in scattered southern areas (Figure 3, Figure 4). Significant changes were spatially limited.

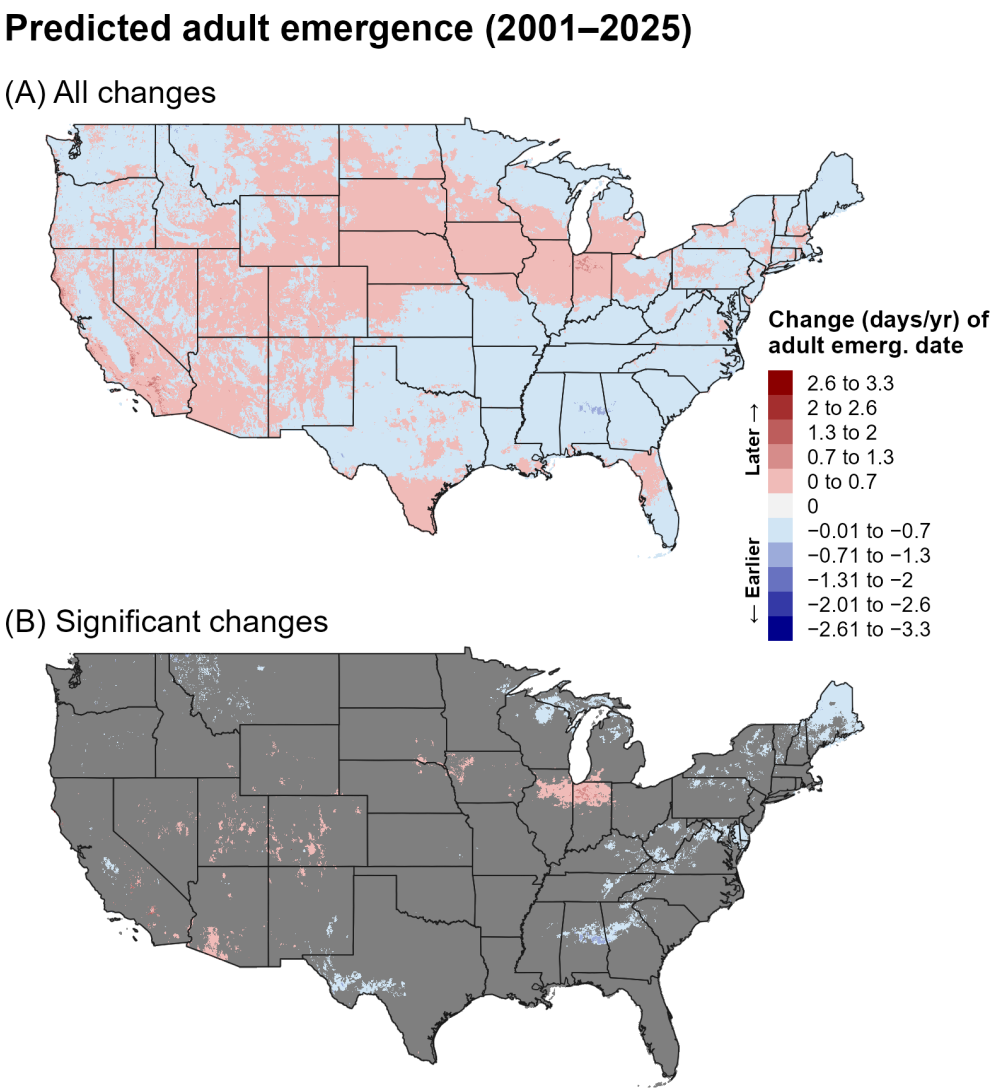


Figure 3. Changes (days/year) in predicted dates of first adult emergence for SLI across the CONUS over the recent 24-year period (2001–2025). Maps show (A) all changes and (B) statistically significant changes ($P < 0.1$); areas shown in gray are not statistically significant.

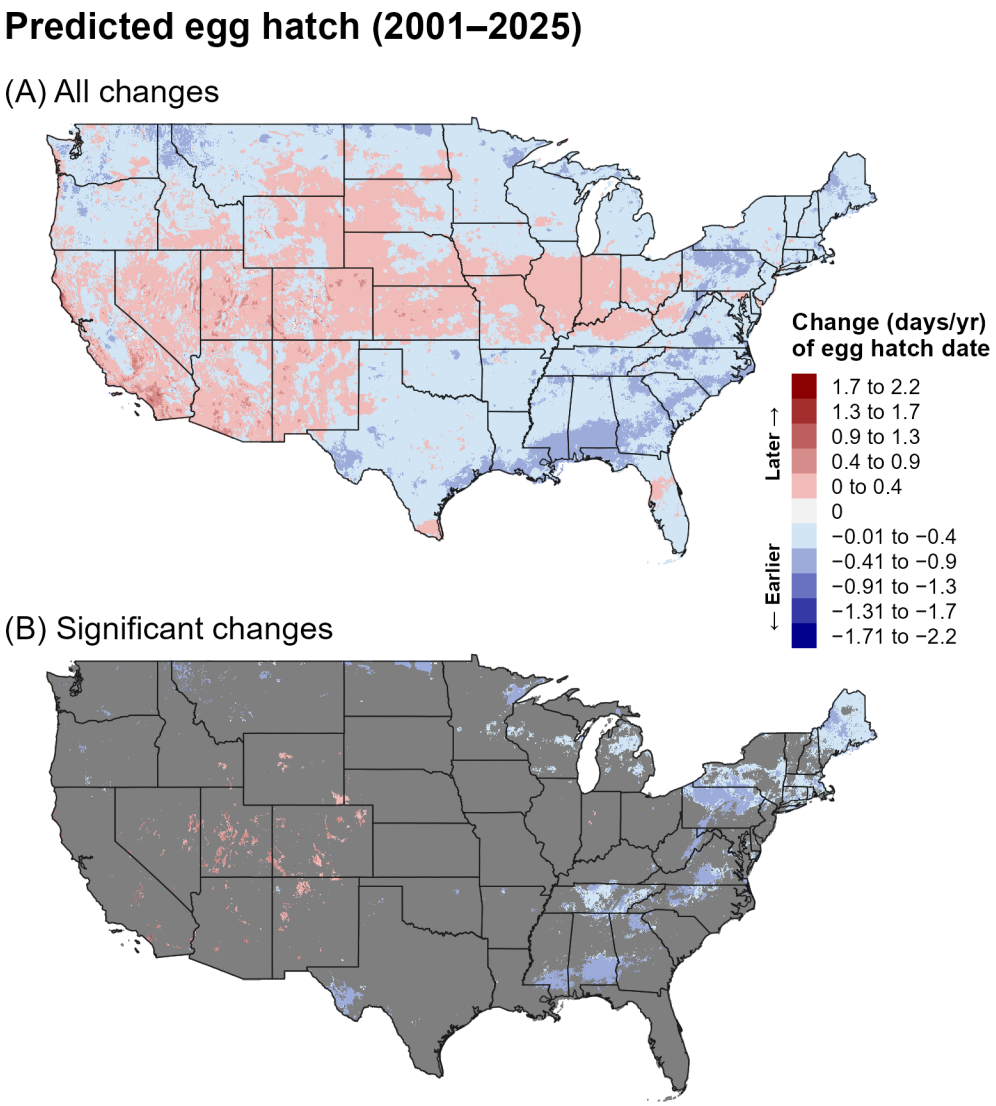


Figure 4. Changes (days/year) in predicted dates of first egg hatch for SLI across the CONUS over the recent 24-year period (2001–2025). Maps show (A) all changes and (B) statistically significant changes ($P < 0.1$); areas shown in gray are not statistically significant.

ii. Climatic suitability

Entire period: 1981–2025

Predicted cold stress decreased across much of the northern United States over 1981–2025, while heat stress changes were highly localized and concentrated mainly in portions of the Southwest (Figure 5, Figure 6). Predicted climatic suitability showed very little statistically significant change (Figure 7).

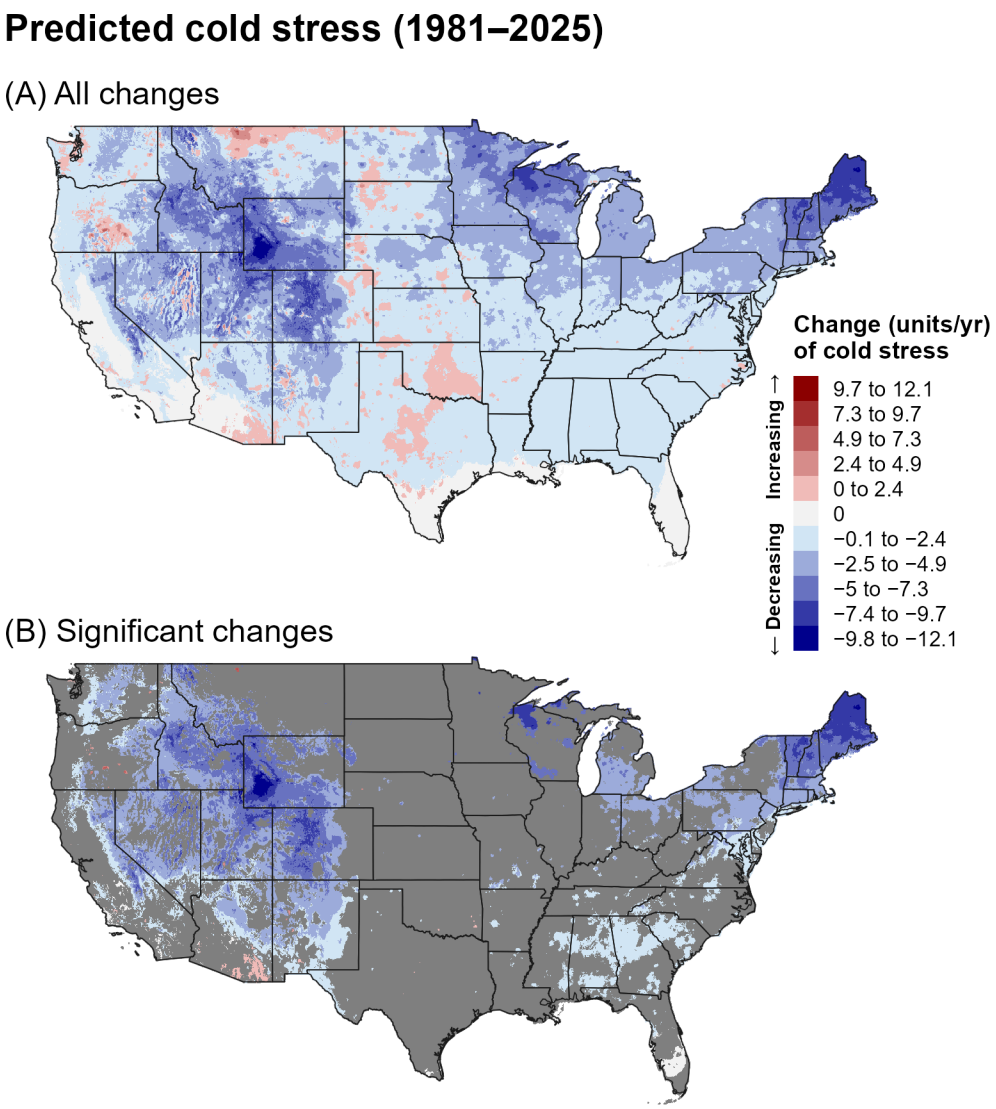


Figure 5. Changes (units/year) in predicted cold stress for SLI across the CONUS over the entire 44-year period (1981–2025). Maps show (A) all changes and (B) statistically significant changes ($P < 0.1$); areas shown in gray are not statistically significant. Areas shown as 0 in (B) had significant trends but very small changes.

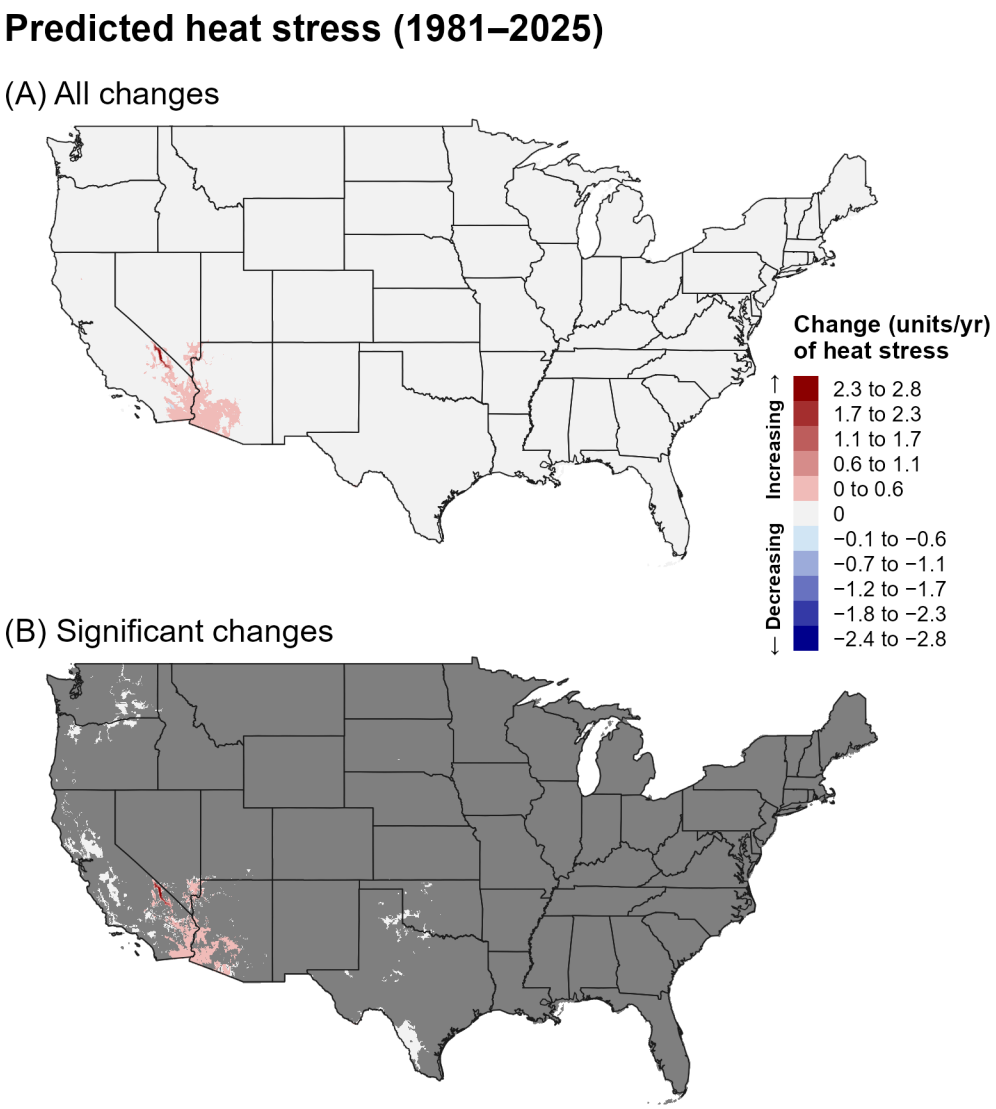


Figure 6. Changes (units/year) in predicted heat stress for SLI across the CONUS over the entire 44-year period (1981–2025). Maps show (A) all changes and (B) statistically significant changes ($P < 0.1$); areas shown in gray are not statistically significant. Areas shown as 0 in (B) had significant trends but very small changes.

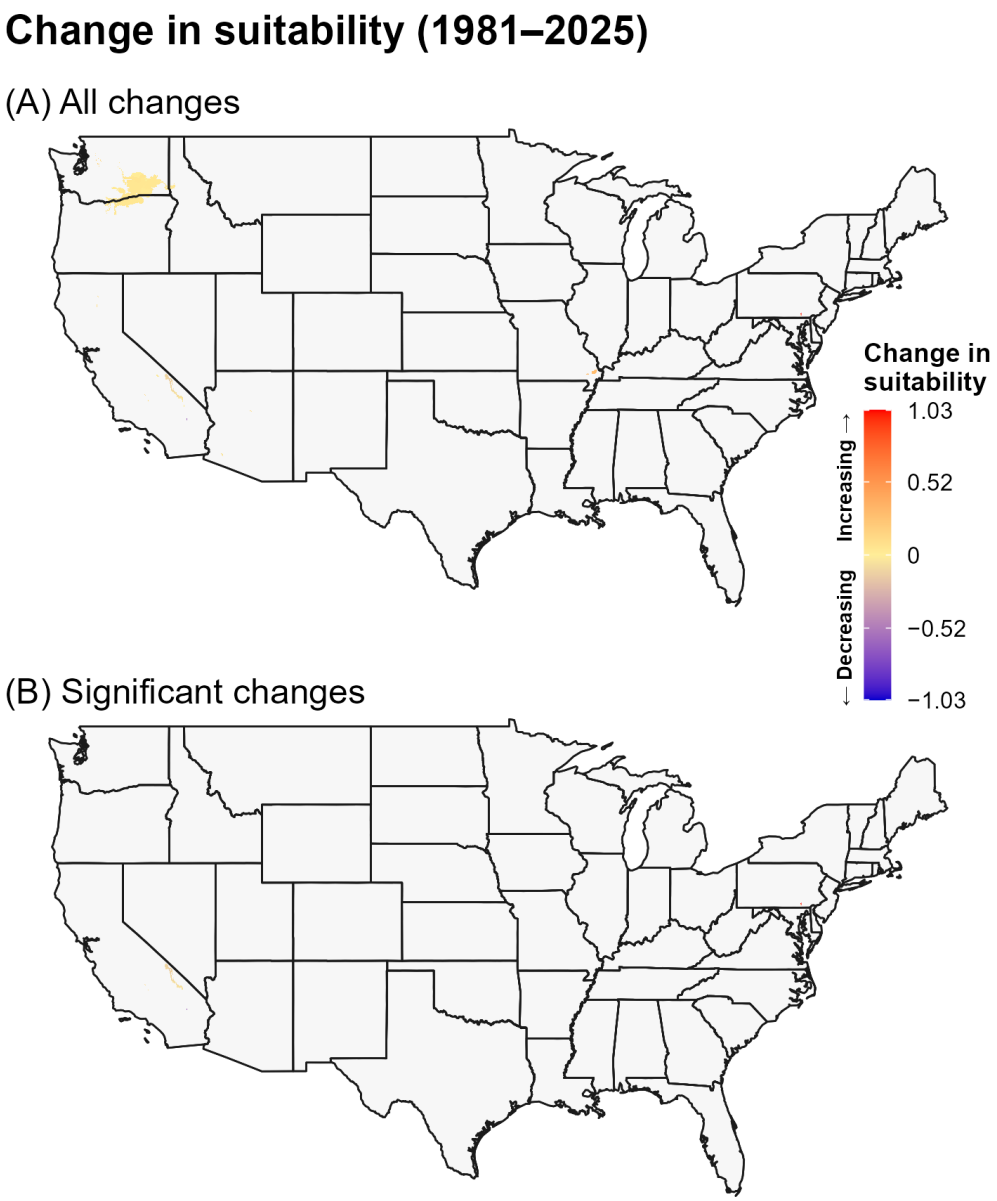


Figure 7. Changes in predicted climatic suitability for SLI across the CONUS over the 44-year period (1981–2025). Suitability is defined as areas not excluded by climate stress. Positive changes (warmer colors) indicate an increased likelihood of establishment, whereas negative changes (cooler colors) indicate a decreased likelihood of establishment.

Recent period: 2001–2025

During 2001–2025, cold-stress trends became more mixed, with increases across portions of the northern and central interior and decreases in scattered areas (Figure 8). Heat-stress changes remained localized to the Southwest (Figure 9). Climatic-suitability changes were essentially absent at the $P < 0.1$ level (Figure 10).

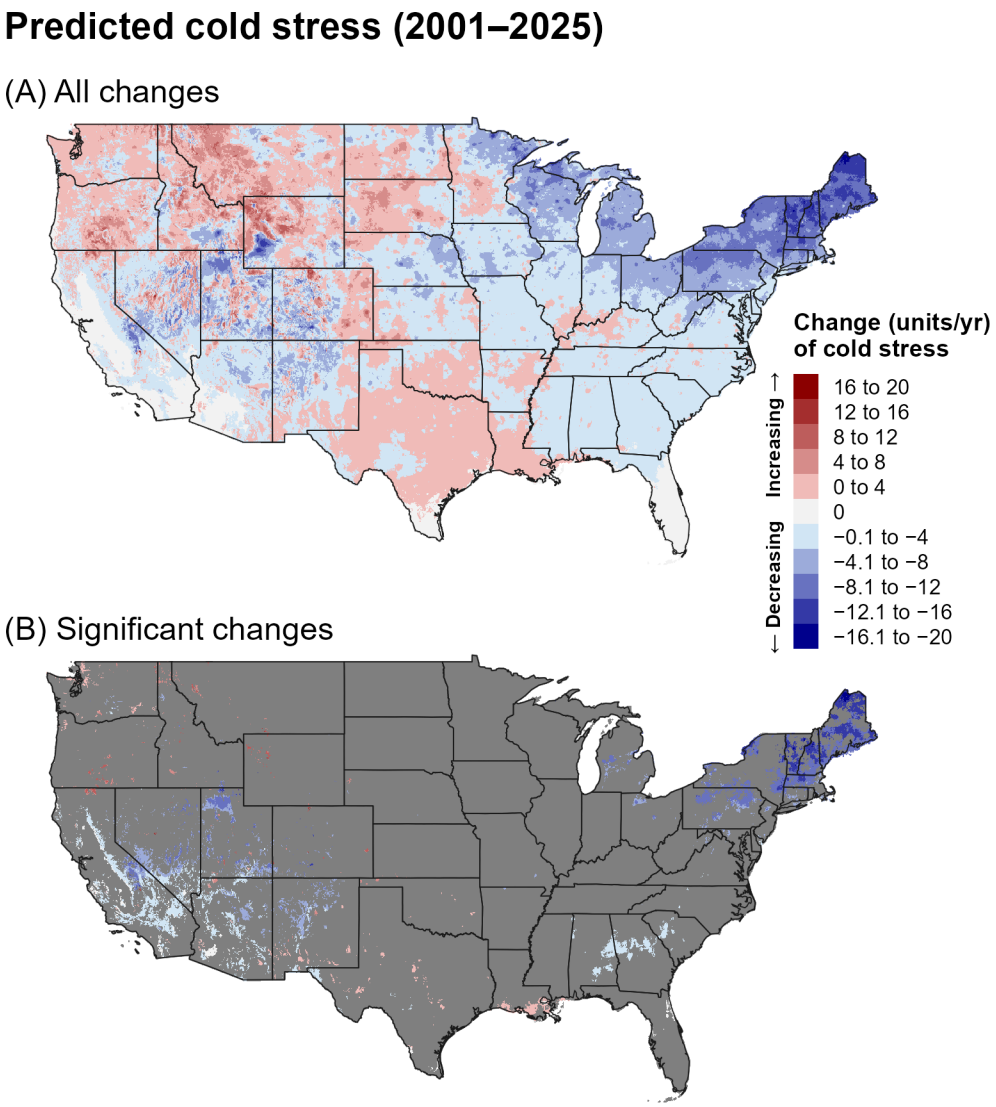


Figure 8. Changes (units/year) in predicted cold stress for SLI across the CONUS over the recent 24-year period (2001–2025). Maps show (A) all changes and (B) statistically significant changes ($P < 0.1$); areas shown in gray are not statistically significant. Areas shown as 0 in (B) had significant trends but very small changes.

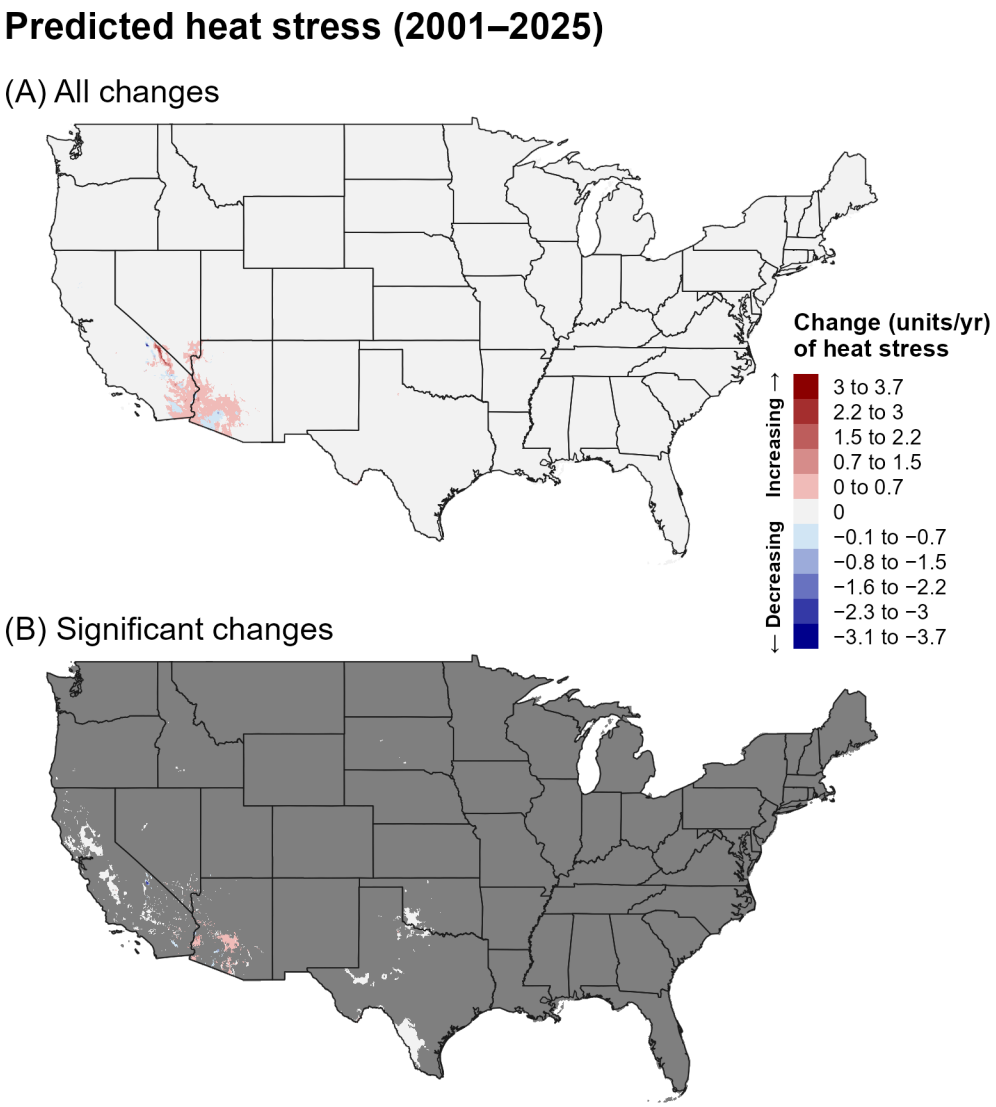


Figure 9. Changes (units/year) in predicted heat stress for SLI across the CONUS over the recent 24-year period (2001–2025). Maps show (A) all changes and (B) statistically significant changes ($P < 0.1$); areas shown in gray are not statistically significant. Areas shown as 0 in (B) had significant trends but very small changes.

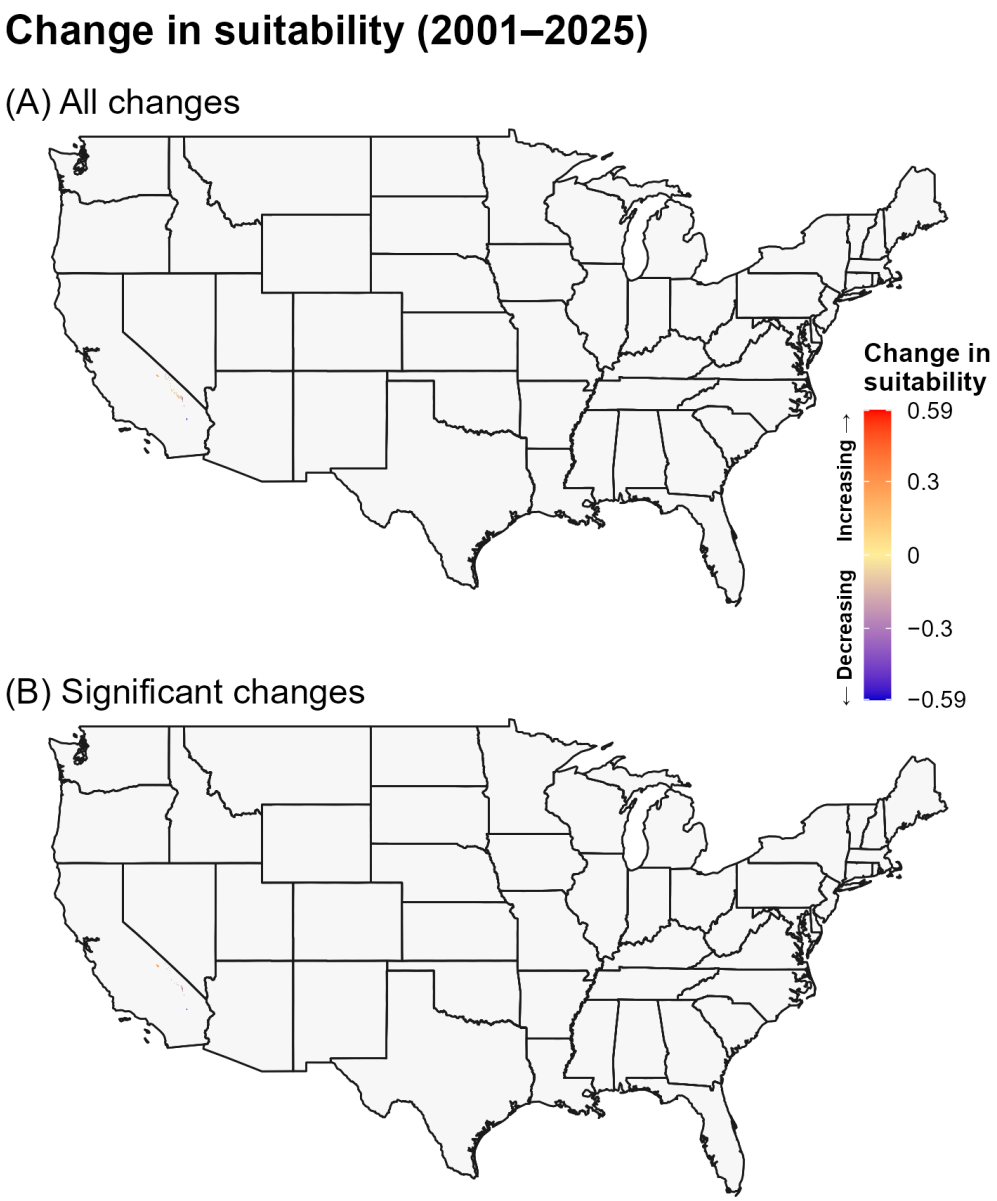


Figure 10. Changes in predicted climatic suitability for SLI across the CONUS over the recent 24-year period (2001–2025). Suitability is defined as areas not excluded by climate stress. Positive changes (warmer colors) indicate an increased likelihood of establishment, whereas negative changes (cooler colors) indicate a decreased likelihood of establishment.

14C. Discussion

The SLI maps indicate that phenological trends vary by event and region, with no single directional pattern dominating the CONUS. This variability suggests that changes in one developmental event should not necessarily be assumed to represent changes in all stages. Where significant phenological shifts occur, monitoring and management schedules may need to be adjusted regionally.

Despite substantial long-term declines in predicted cold stress across northern regions, the climatic-suitability maps showed little significant change. The limited heat-stress signal further suggests that climatic conditions remain broadly within the modeled suitability range across most of the CONUS, at least with respect to the

stress thresholds represented by the model.

The stronger and more spatially extensive trends in predicted phenology and climate stress over the full 44-year period may reflect both the larger sample size, which can reduce noise and improve detection of directional patterns, and differences in the underlying climate trends between the two time periods. In particular, the recent 24-year period may capture climate variability or regional trends that differ from those observed over the longer 44-year period. Thus, differences between the two periods should be interpreted as reflecting both long-term climate trends and shorter-term regional variability.

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